

Lecture 19: Waveform tomography

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Outline and Motivation

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- It fully accounts for the complex physics of wave propagation by using numerical solutions to the elastic wave equation.
- It treats the inverse problem as fundamentally **non-linear**.
- Our focus will be on how to solve this large-scale inverse problem using **gradient-based optimisation**.

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Least-Squares Misfit Function

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- The value of the misfit J depends on the synthetic wavefield $\mathbf{u}$, which in turn depends on the underlying Earth model parameters, $\mathbf{m}$. We can therefore think of the misfit as a function $J(\mathbf{m})$.
- **The goal of waveform tomography is to find the model $\mathbf{m}$ that minimises this misfit function.**

Solving the Inverse Problem: Optimisation

We want to find a model $\overline{\mathbf{m}}$ that is a minimum of the misfit function $J(\mathbf{m})$. This occurs where the gradient (or functional derivative) of the misfit is zero:

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- We must use **local, gradient-based optimisation methods**. These iteratively improve an initial model by moving it in a direction related to the negative gradient.
- A major drawback is that these methods can get stuck in a **local minimum** of the misfit function, which may not be the true global minimum.

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- 4 Find the optimal step length λ using a line search to achieve the largest misfit reduction.
- 5 Repeat until the gradient is nearly zero or the misfit stops decreasing.

Calculating the Gradient

The key challenge is calculating the gradient $DJ(\mathbf{m})$ efficiently.

A Bad Idea: Finite Differences

We could approximate the derivative by perturbing each model parameter one by one and running a new simulation for each.

$$\frac{\partial J}{\partial m_i} \approx \frac{J(\mathbf{m} + \epsilon \mathbf{e}_i) - J(\mathbf{m})}{\epsilon}$$

This would require $N + 1$ wavefield simulations, where N is the number of model parameters. Since N can be millions, this is computationally prohibitive.

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A Good Idea: The Adjoint Method

There is a much more efficient way to calculate the exact gradient, known as the **adjoint method**.

The Adjoint Method (1/2): Concept

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- It introduces an **adjoint wavefield**, u' . This adjoint field satisfies the same wave equation as the normal "forward" wavefield.
- However, the source for the adjoint wavefield is the **data residual** ($u - u^{obs}$), but applied as a force at the *receiver* location and running **backwards in time**.

The Adjoint Method (2/2): The Payoff

Calculating the Sensitivity Kernel

The gradient of the misfit with respect to any model parameter can be calculated by combining the forward field ($\mathbf{u}$) and the adjoint field ($\mathbf{u}'$). For example, the sensitivity to density, ρ , is:

$$K_{\rho} = \int_0^T \frac{\partial \mathbf{u}}{\partial t} \cdot \frac{\partial \mathbf{u}'}{\partial t} dt$$

Similar expressions exist for elastic parameters.

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Computational Advantage

The total cost to compute the gradient for **all** model parameters is:

- ➊ One forward simulation to get $\mathbf{u}$ and the data residual.
- ➋ One backward (adjoint) simulation to get $\mathbf{u}'$.
- ➌ One step to combine the saved fields.

The cost is independent of the number of model parameters, making

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*This shape – broad around the ray path but hollow in the middle – led to the name “**banana-doughnut kernel**”.*

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- The gradient of the misfit function is calculated efficiently using the **adjoint method**, which involves one forward and one backward (adjoint) wave simulation.
- The cost of the adjoint method is independent of the number of model parameters, making large-scale problems computationally tractable.
- Full-wave sensitivity kernels (**banana-doughnut kernels**) have finite width related to the Fresnel zone, differing from the infinitely thin sensitivities of ray theory.